

3.3 Sub-module 1: Getting Familiar with Data Visualization (DV)

This sub-module teaches students to apply Computer Vision (CV) algorithms to analyse and visualise data [DV-Obj]. Students develop procedural knowledge through practical guided activities (e.g. Midjourney step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating an infographic). Students are first provided with a basic literacy of DV algorithms and data visualisation (DV) techniques including technical vocabulary (DV-Act1) and primary functionalities (DV-Act2). This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the disciplines (CV and DV) and their interrelationship in analysing and visualising complex phenomena (Omena, 2021). Once basic literacy is introduced, specific case studies are provided to contextualise the application of CV within the design research (DV-Act3).

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial (DV-Act4.3) on how to analyse data with CV algorithms (e.g. Meme Spector) and visualise the results (e.g. Gephi) starting from setting a research question (DV-Act4.1) and collecting and organising data related to the phenomena (DV-Act4.2). Then, to familiarise themselves with the procedure, students individually apply what they learned by investigating the dataset to answer the research question (DV-Act5). Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results (DV-Act6). This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others (DV-Act7).

Educational Objective {DV-Obj} :

Students will learn to apply Computer Vision algorithms for analyze and visualize data.

Sub-module 1: Getting familiar w/ DV	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{DV-Act.1}				
(b) Conceptual Knowledge		{DV-Act.2} {DV-Act.3}				
(c) Procedural Knowledge		{DV-Act.4} {DV-Act.4.1, 4.2, 4.3}	{DV-Obj} {DV-Act.5}			
(d) Meta Knowledge				{DV-Act.6}	{DV-Act.7}	

- {DV-Act1} Activities intended to provide students with vocabulary knowledge of Computer Vision;
- {DV-Act2} Activities intended to provide students with a basic functionality of DV Algorithms and DV techniques;
- {DV-Act3} Activities intended to provide students with practical examples of computer vision applied to design research;
- {DV-Act4} Activities intended to explain the procedure of analyzing data with computer vision algorithms;
- {DV-Act4.1} Activities intended to introduce students in familiarising, collecting or creating data;
- {DV-Act4.2} Activities intended to set or aligning with a research question;
- {DV-Act4.3} Activities intended to provide a step-by-step tool tutorial, such as Memespecotor, Gephi;
- {DV-Act5} Activities intended to allow students to individually analyzes data with computer vision for a research goal;
- {DV-Act6} Activities intended to provide students with a structured template for process and results documentation;
- {DV-Act7} Activities intended to make students explaining and questioning the process.